

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive")

```
import pandas as pd

path = "/content/drive/MyDrive/train.csv"
df = pd.read_csv(path)
```

```
#1st five rows
df.head()
```

	sample_id	catalog_content	image_link	price	
0	33127	Item Name: La Victoria Green Taco Sauce Mild, ...	https://m.media-amazon.com/images/I/51mo8htwTH...	4.89	
1	198967	Item Name: Salerno Cookies, The Original Butte...	https://m.media-amazon.com/images/I/71YtriHAA...	13.12	
2	261251	Item Name: Bear Creek Hearty Soup Bowl, Creamy...	https://m.media-amazon.com/images/I/51+PFEEw-...	1.97	
3	55858	Item Name: Judee's Blue Cheese Powder 11.25 oz...	https://m.media-amazon.com/images/I/41mu0HAToD...	30.34	
4	292686	Item Name: kedem Sherry Cooking Wine, 12.7 Oun...	https://m.media-amazon.com/images/I/41sA037+Qv...	66.49	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
#last 5 rows
df.tail()
```

	sample_id	catalog_content	image_link	price	
74995	41424	Item Name: ICE BREAKERS Spearmint Sugar Free M...	https://m.media-amazon.com/images/I/81p9PcPsff...	10.395	
74996	35537	Item Name: Davidson's Organics, Vanilla Essenc...	https://m.media-amazon.com/images/I/51DDKoa+mb...	35.920	
74997	249971	Item Name: Jolly Rancher Hard Candy - Blue Ras...	https://m.media-amazon.com/images/I/91R2XCcpUf...	50.330	
74998	188322	Item Name: Nescafe Dolce Gusto Capsules - CARA...	https://m.media-amazon.com/images/I/51W40YU98+...	15.275	
74999	298504	Item Name: Pimenton de la Vera - Picante (2.47...	https://m.media-amazon.com/images/I/81dFnP6C4...	28.240	

```
#size
df.shape
```

```
(75000, 4)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 75000 entries, 0 to 74999
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sample_id       75000 non-null  int64
1   catalog_content 75000 non-null  object
2   image_link      75000 non-null  object
3   price           75000 non-null  float64
dtypes: float64(1), int64(1), object(2)
memory usage: 2.3+ MB
```

```
df.isnull()
```

	sample_id	catalog_content	image_link	price	
0	False	False	False	False	
1	False	False	False	False	
2	False	False	False	False	
3	False	False	False	False	
4	False	False	False	False	
...	...	...	...	...	
74995	False	False	False	False	
74996	False	False	False	False	
74997	False	False	False	False	
74998	False	False	False	False	
74999	False	False	False	False	

75000 rows × 4 columns

```
#checking null values
df.isnull().sum()
```

```

0
sample_id    0
catalog_content  0
image_link    0
price        0

dtype: int64
```

```
#checking duplicates
df.duplicated().sum()
```

```
np.int64(0)
```

```
#finding garbage values
for i in df.select_dtypes(include="object").columns:
    print(df[i].value_counts())
    print("*****10)
```

```
catalog_content
Item Name: PAPYRUS Everyday Card, 1 EA\nValue: 1.0\nUnit: Count\n
Item Name: French's White Cheddar Crispy Fried Onions, 6 oz\nBullet Point 1: Made with real onions\nBull
Item Name: Rose's Sweetened Grenadine Syrup 25oz Bottle | Perfect for Cocktails, Beverages, and Mixers\r
Item Name: PAPYRUS Everyday Card, 1 EA\nProduct Description: Lifestyle\nValue: 1.0\nUnit: Count\n
```

```
Item Name: Eden Foods Salt Gomasio Ssme Seawd\nValue: 1.0\nUnit: Count\n
```

```
Item Name: TEDDY GRAHAMS Honey Graham Snacks, 40 Kids Snack Packs (4 Boxes)\nBullet Point 1: Four boxes
Item Name: Sour Blue Raspberry Licorice Bits by Its Delish, 2 lbs\nBullet Point 1: Delicious Sweet and S
Item Name: Krusteaz Baking Mix, Pumpkin Spice Quick Bread Mix, Made with Real Pumpkin & No Artificial Fl
Item Name: Jack Link's Beef Jerky, Sweet & Hot Flavor, 2.85 oz. - Flavorful Meat Snack, 9g of Protein ar
Item Name: Grand Parade Coffee, 3 Lbs Kenya AA Rainforest Alliance Unroasted Green Coffee Beans, Special
```

```
Name: count, Length: 74900, dtype: int64
```

```
*****
```

```
image_link
```

```
https://m.media-amazon.com/images/I/51m1gdQJW2L.jpg 52
```

```
https://m.media-amazon.com/images/I/71LRdXdqc0L.jpg 20
```

```
https://m.media-amazon.com/images/I/21mMXLWiDOL.jpg 13
```

```
https://m.media-amazon.com/images/I/61md5v6UPNL.jpg 12
```

```
https://m.media-amazon.com/images/I/71FMI9t03HL.jpg 12
```

```
..
```

```
https://m.media-amazon.com/images/I/513Bo+IWjSL.jpg 1
```

```
https://m.media-amazon.com/images/I/91psQbgJuxL.jpg 1
```

```
https://m.media-amazon.com/images/I/81ukU3dmKsL.jpg 1
```

```
https://m.media-amazon.com/images/I/912+ZelgZVL.jpg 1
```

```
https://m.media-amazon.com/images/I/81Ly1kSbjVL.jpg 1
```

```
Name: count, Length: 72288, dtype: int64
```

```
*****
```

```
#cleaning text by removing \n and multiple space, convert to lower and trim
```

```
import pandas as pd
```

```
import re
```

```
def safe_clean(text):
```

```
    text = text.lower()
```

```
    text = re.sub(r'\n+', ' ', text)
```

```
    text = re.sub(r'\s+', ' ', text)
```

```
df['clean_text'] = df['catalog_content'].apply(safe_clean)
```

```
#finding garbage values
```

```
for i in df.select_dtypes(include="object").columns:
```

```
    print(df[i].value_counts())
```

```
    print("****10)
```

```
catalog_content
```

```
Item Name: PAPYRUS Everyday Card, 1 EA\nValue: 1.0\nUnit: Count\n
```

```
Item Name: French's White Cheddar Crispy Fried Onions, 6 oz\nBullet Point 1: Made with real onions\nBull
```

```
Item Name: Rose's Sweetened Grenadine Syrup 25oz Bottle | Perfect for Cocktails, Beverages, and Mixers\r
```

```
Item Name: PAPYRUS Everyday Card, 1 EA\nProduct Description: Lifestyle\nValue: 1.0\nUnit: Count\n
```

```
Item Name: Eden Foods Salt Gomasio Ssme Seawd\nValue: 1.0\nUnit: Count\n
```

```
Item Name: TEDDY GRAHAMS Honey Graham Snacks, 40 Kids Snack Packs (4 Boxes)\nBullet Point 1: Four boxes
```

```
Item Name: Sour Blue Raspberry Licorice Bits by Its Delish, 2 lbs\nBullet Point 1: Delicious Sweet and S
```

```
Item Name: Krusteaz Baking Mix, Pumpkin Spice Quick Bread Mix, Made with Real Pumpkin & No Artificial Fl
```

```
Item Name: Jack Link's Beef Jerky, Sweet & Hot Flavor, 2.85 oz. - Flavorful Meat Snack, 9g of Protein ar
```

```
Item Name: Grand Parade Coffee, 3 Lbs Kenya AA Rainforest Alliance Unroasted Green Coffee Beans, Special
```

```
Name: count, Length: 74900, dtype: int64
```

```
*****
```

```
image_link
```

```
https://m.media-amazon.com/images/I/51m1gdQJW2L.jpg 52
```

```
https://m.media-amazon.com/images/I/71LRdXdqc0L.jpg 20
```

```
https://m.media-amazon.com/images/I/21mMXLWiDOL.jpg 13
```

```
https://m.media-amazon.com/images/I/61md5v6UPNL.jpg 12
```

```
https://m.media-amazon.com/images/I/71FMI9t03HL.jpg 12
```

```
..
```

```
https://m.media-amazon.com/images/I/513Bo+IWjSL.jpg 1
```

```
https://m.media-amazon.com/images/I/91psQbgJuxL.jpg 1
```

```
https://m.media-amazon.com/images/I/81ukU3dmKsL.jpg 1
```

```
https://m.media-amazon.com/images/I/912+ZelgZVL.jpg 1
```

```
https://m.media-amazon.com/images/I/81Ly1kSbjVL.jpg 1
```

```
Name: count, Length: 72288, dtype: int64
```

```
*****
```

```
Series([], Name: count, dtype: int64)
```

```
*****
```

df.head()

1 to 5 of 5 entries

Filter



index	sample_id	catalog_content	image_link	price	clean_text
0	33127	Item Name: La Victoria Green Taco Sauce Mild, 12 Ounce (Pack of 6) Value: 72.0 Unit: Fl Oz	https://m.media-amazon.com/images/I/51mo8htwTHL.jpg	4.89	null
1	198967	Item Name: Salerno Cookies, The Original Butter Cookies, 8 Ounce (Pack of 4) Bullet Point 1: Original Butter Cookies: Classic butter cookies made with real butter Bullet Point 2: Variety Pack: Includes 4 boxes with 32 cookies total Bullet Point 3: Occasion Perfect: Delicious cookies for birthdays, weddings, anniversaries Bullet Point 4: Shareable Treats: Fun to give and enjoy with friends and family Bullet Point 5: Salerno Brand: Trusted brand of delicious butter cookies since 1925 Value: 32.0 Unit: Ounce	https://m.media-amazon.com/images/I/71YtriIHAAL.jpg	13.12	null
2	261251	Item Name: Bear Creek Hearty Soup Bowl, Creamy Chicken with Rice, 1.9 Ounce (Pack of 6) Bullet Point 1: Loaded with hearty long grain wild rice and vegetables Bullet Point 2: Full of hearty goodness Bullet Point 3: Single serve bowls Bullet Point 4: Easy to prepare mix Bullet Point 5: 0 grams trans fat Value: 11.4 Unit: Ounce	https://m.media-amazon.com/images/I/51+PFEE-w-L.jpg	1.97	null
3	55858	Item Name: Judee's Blue Cheese Powder 11.25 oz - Gluten-Free and Nut-Free - Use in Seasonings and Salad Dressings - Great for Dips, Spreads and Sauces - Made in USA Bullet Point 1: Add to your favorite appetizers, dips & spreads. Use to season popcorn or warmed pita chips. Bullet Point 2: Sprinkle over french fries, fried chicken, mashed potatoes, roasted veggies, pasta, and more Bullet Point 3: Made in a dedicated gluten-free facility and shipped in a standup, resealable pouch to ensure freshness Bullet Point 4: Ingredients: Blue Cheese (Milk, Salt, Cultures, & Enzymes) and Disodium Phosphate Bullet Point 5: Since 2009, Judee's has been dedicated to providing fresh, allergy-conscious ingredients, great for your recipes and even better for your family Product Description: Judees Powdered Blue Cheese cheddar cheese powder is an alternative to mozzarella cheese shredded or american cheese slices deli. Make your own alfredo sauce with heavy cream and black buffalo dip with this powder. It adds extra flavor to salad dressing like ranch dressing and great on pizza dough or cauliflower pasta. Add to macaroni and cheese or popcorn seasoning for more aroma and cheesy feel. Combine with mustard and other ingredients to create your own dressing for buffalo chicken or buffalo wings. Value: 11.25 Unit: Ounce	https://m.media-amazon.com/images/I/41mu0HAToDL.jpg	30.34	null
4	292686	Item Name: kedem Sherry Cooking Wine, 12.7 Ounce - 12 per case. Bullet Point: kedem Sherry Cooking Wine, 12.7 Ounce - 12 per case. Value: 12.0 Unit: Count	https://m.media-amazon.com/images/I/41sA037+QvL.jpg	66.49	null

df.describe()

	sample_id	price	
count	75000.000000	75000.000000	
mean	149841.917707	23.647654	
std	86585.346513	33.376932	
min	0.000000	0.130000	
25%	73845.750000	6.795000	
50%	150129.000000	14.000000	
75%	225040.250000	28.625000	
max	299438.000000	2796.000000	

```
df.describe(include="object")
```

	catalog_content	image_link	clean_text	
count	75000	75000	0	
unique	74900	72288	0	
top	Item Name: PAPYRUS Everyday Card, 1 EA\nValue:...	https://m.media-amazon.com/images/I/51m1gdQJW2...	NaN	
freq	5	52	NaN	

```
def safe_clean(text):
    if not isinstance(text, str):
        return ""
    text = text.lower()
    text = re.sub(r'\n+', ' ', text)
    text = re.sub(r'\s+', ' ', text)
    return text.strip()

df['clean_text'] = df['catalog_content'].astype(str).apply(safe_clean)
print(df['clean_text'].head())
```

```
0    item name: la victoria green taco sauce mild, ...
1    item name: salerno cookies, the original butte...
2    item name: bear creek hearty soup bowl, creamy...
3    item name: judee's blue cheese powder 11.25 oz...
4    item name: kedem sherry cooking wine, 12.7 oun...
Name: clean_text, dtype: object
```

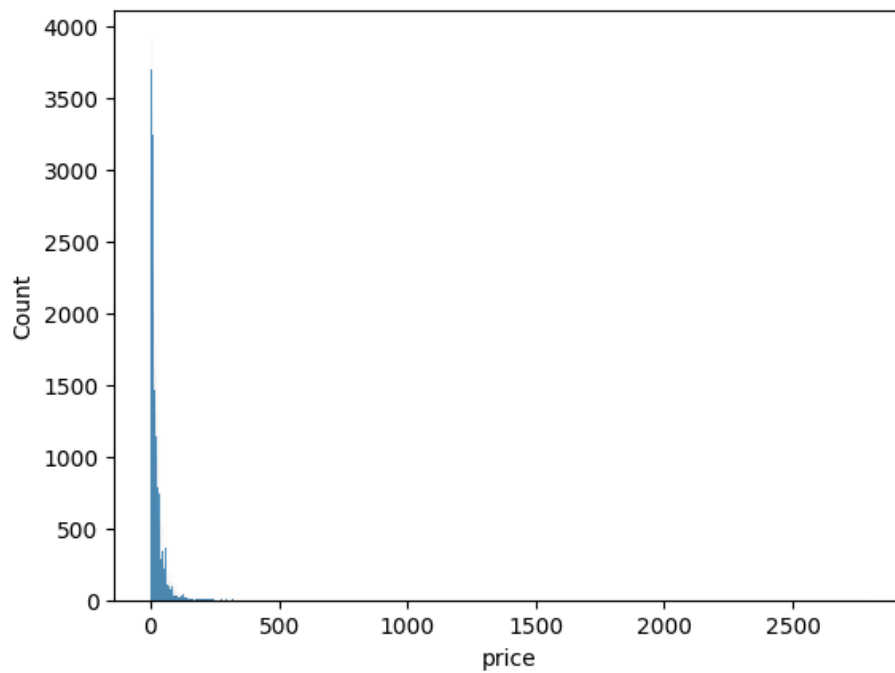
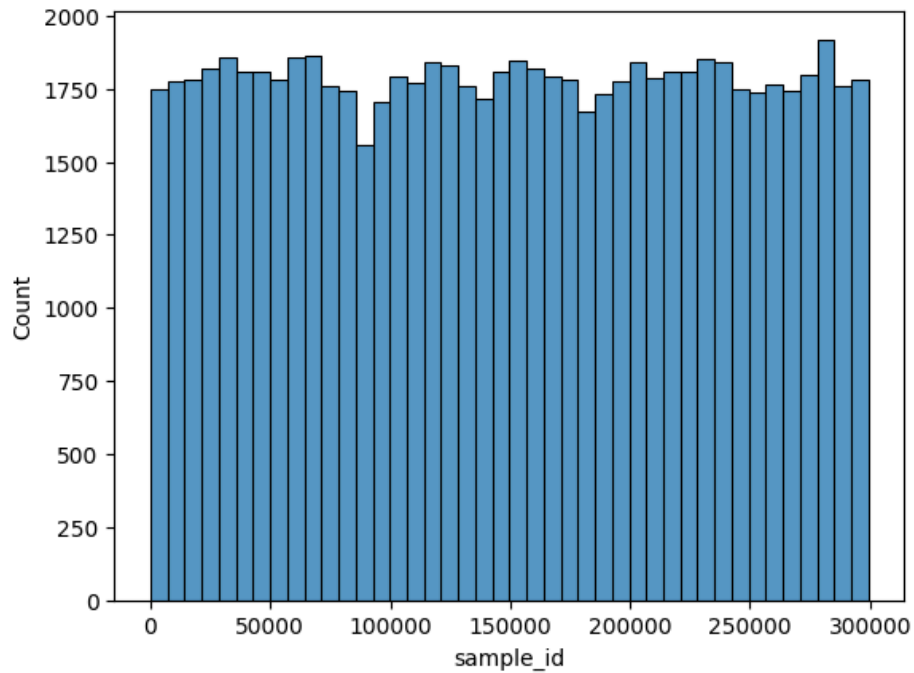
```
df.describe(include="object")
```

	catalog_content	image_link	clean_text	
count	75000	75000	75000	
unique	74900	72288	74893	
top	Item Name: PAPYRUS Everyday Card, 1 EA\nValue:...	https://m.media-amazon.com/images/I/51m1gdQJW2...	item name: papyrus everyday card, 1 ea value: ...	
freq	5	52	5	

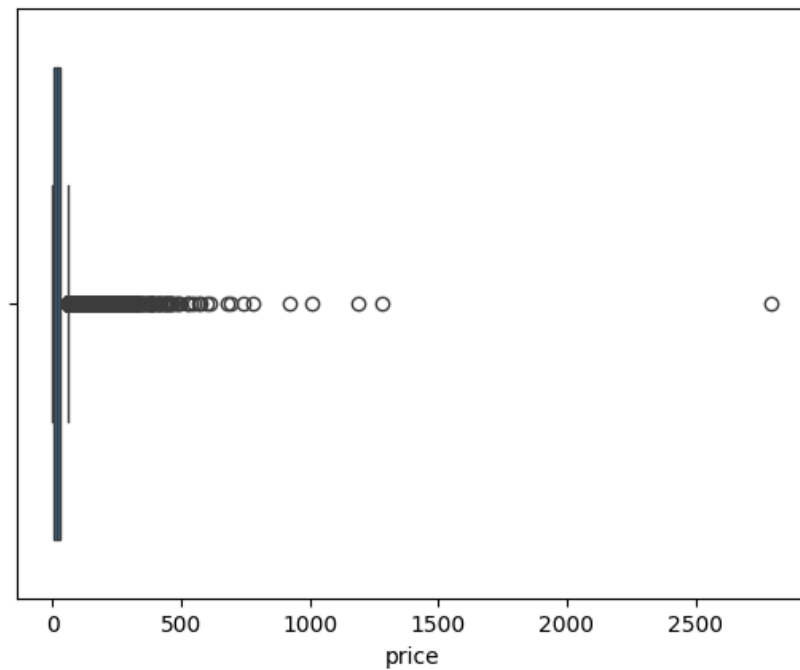
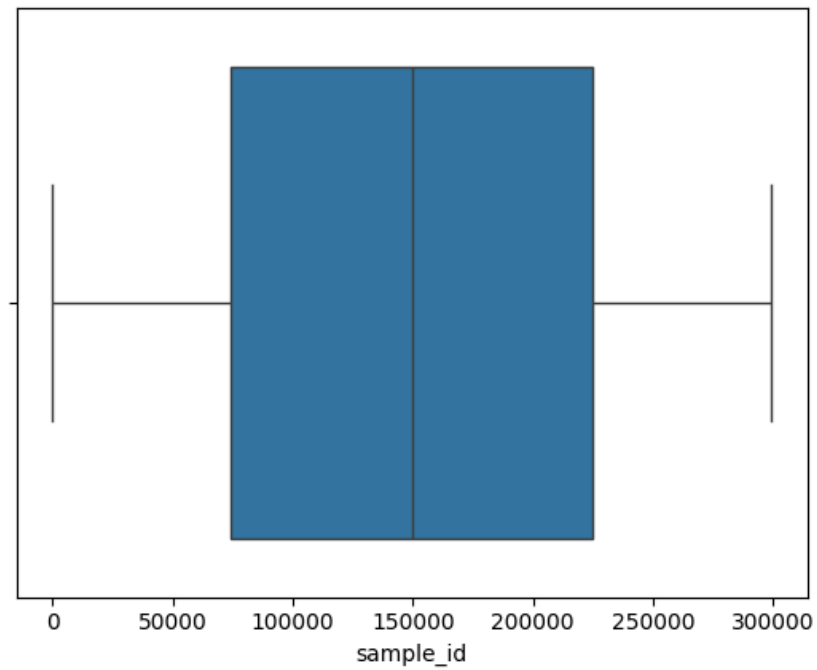
```
# understanding distribution by plotting histogram
import seaborn as sns
import matplotlib.pyplot as plt

import warnings
warnings.filterwarnings("ignore")
```

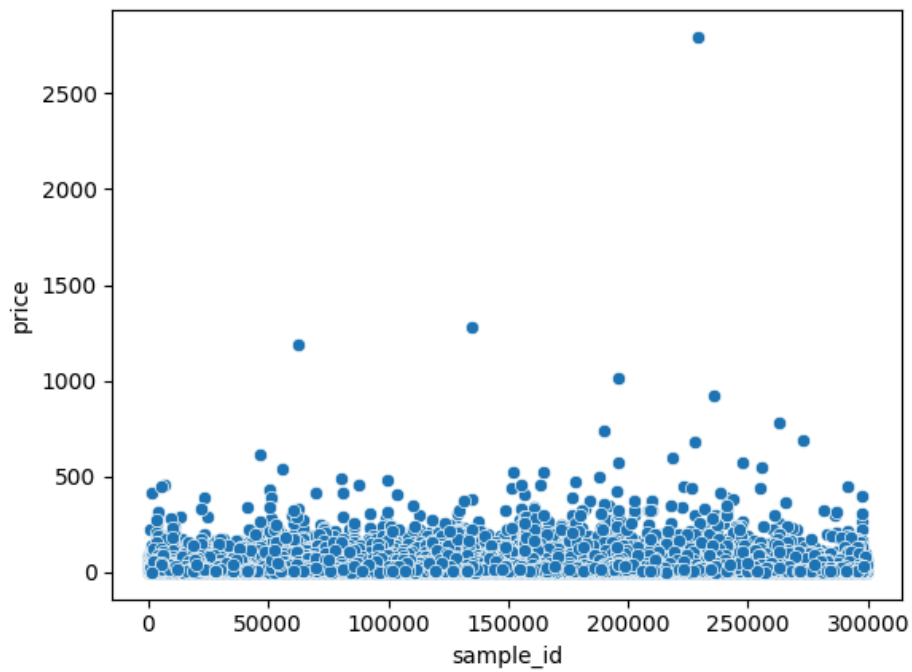
```
for i in df.select_dtypes(include="number").columns:  
    sns.histplot(data=df,x=i)  
    plt.show()
```



```
#boxplot to identify outliers  
import seaborn as sns  
import matplotlib.pyplot as plt  
  
import warnings  
warnings.filterwarnings("ignore")  
for i in df.select_dtypes(include="number").columns:  
    sns.boxplot(data=df,x=i)  
    plt.show()
```



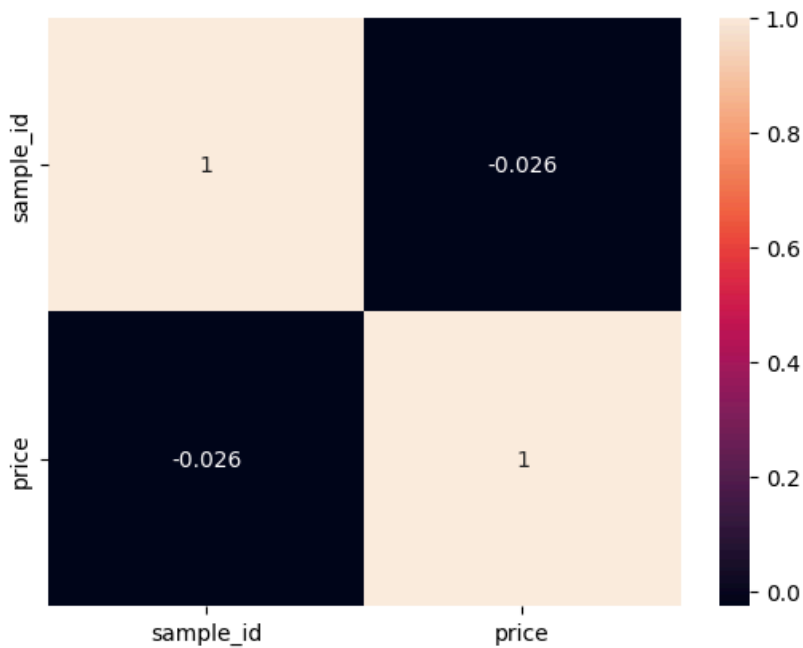
```
#scatter plot to understand relationship
for i in ['sample_id']:
    sns.scatterplot(data=df,x=i,y='price')
plt.show()
```



```
s=df.select_dtypes(include="number").corr()
```

```
sns.heatmap(s,annot=True)
```

<Axes: >



```
#Handling missing values  
df.isnull().sum()
```


	0
sample_id	0
catalog_content	0
image_link	0
price	0
clean_text	0

dtype: int64

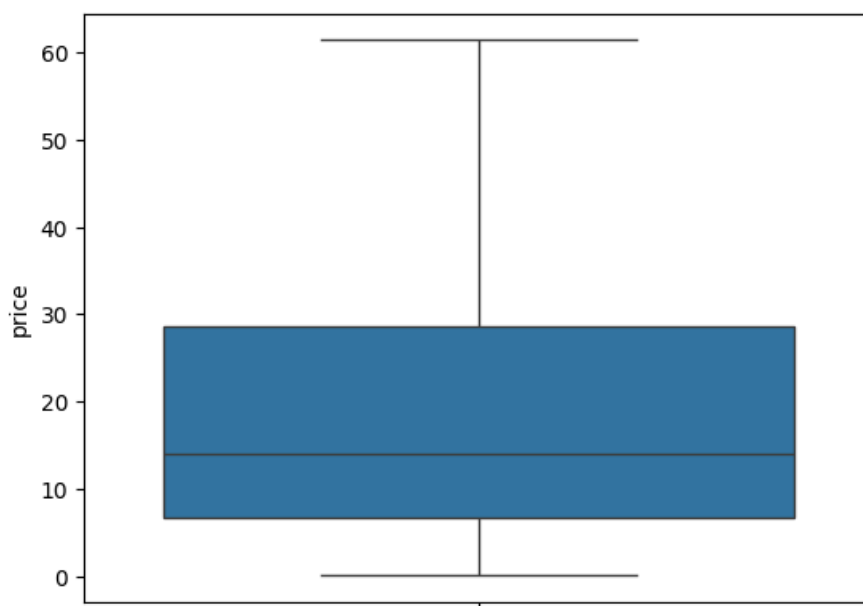
```
#Handling outliers
def wisker(col):
    q1 = df['price'].quantile(0.25)
    q3 = df['price'].quantile(0.75)
    iqr = q3 - q1
    lw=q1-1.5*iqr
    uw=q3+1.5*iqr
    return lw,uw
```

```
wisker(df['price'])
```

```
(np.float64(-25.949999999999996), np.float64(61.37))
```

```
import numpy as np
for i in ['price']:
    lw,uw=wisker(df[i])
    df[i]=np.where(df[i]<lw,lw,df[i])
    df[i]=np.where(df[i]>uw,uw,df[i])
```

```
for i in ['price']:
    sns.boxplot(df[i])
    plt.show()
```



```
#handling duplicates
df.drop_duplicates()
#we do not have any duplicates
```

	sample_id	catalog_content	image_link	price	clean_text	
0	33127	Item Name: La Victoria Green Taco Sauce Mild, ...	https://m.media- amazon.com/images/I/51mo8htwTH...	4.890	item name: la victoria green taco sauce mild, ...	
1	198967	Item Name: Salerno Cookies, The Original Butte...	https://m.media- amazon.com/images/I/71YtrilHAA...	13.120	item name: salerno cookies, the original butte...	
		Item Name: Bear Creek	https://m.media-		item name: bear	